

SCAFFOLD

Business Plan

AI-First Support Agent for Studying, Learning, and Building

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1. Overview of the Entry

1.1 Background

The global AI market is projected to reach USD 1.81 trillion by 2030 (Grand View Research). Within this space, AI-powered productivity tools are among the fastest-growing segments, driven by the rapid adoption of large language models and intelligent agents.

Despite this growth, a fundamental problem persists: most productivity tools still require the user to do the organizing. Users must decide what is a task, what should be a note, what to search, what to remember, and what to do next. Tools designed to help actually add cognitive load.

At the same time, today's AI assistants are stateless. They start every session from scratch, have no memory of past work, no sense of the user's progress, and no ability to follow through. Scaffold was built to solve both problems.

1.2 Project Overview

Scaffold is an AI-first support agent that acts as a persistent, context-aware colleague for studying, learning, and building. Unlike conventional assistants that wait for instructions, Scaffold observes what the user is doing, infers what help is needed, and proactively guides, teaches, challenges, and follows through.

The core idea is simple: the user talks naturally, and the agent handles the rest -- deciding when to create a task, capture a note, search for information, or recommend the next step. Scaffold supports the full lifecycle of real work: exploring, understanding, planning, executing, reviewing, and retaining knowledge.

Scaffold is built on a modern AI agent architecture with real-time streaming, durable workflow execution, hybrid knowledge retrieval (combining document search with knowledge graphs), and an emotional engine that lets the agent maintain a genuine, evolving relationship with the user across sessions.

1.3 Team Introduction

Scaffold is built by a team with deep expertise in AI application development, full-stack engineering, and human-computer interaction. The founding team brings experience in multi-agent system design, production AI deployment, modern web development, and product design for AI-first experiences.

The team operates with a disciplined, layered approach: each feature must be stable and validated in real daily use before the next one begins. This ensures every component is production-ready, not just prototyped.

1.4 Core Technology

Scaffold is built on a production-grade stack designed for reliability, real-time responsiveness, and scalability:

- Backend: High-performance Python API with async processing and real-time streaming
- AI Engine: Stateful multi-agent orchestration with planning, execution, and safety layers

- Knowledge Retrieval: Hybrid system combining semantic document search with knowledge graph queries
- Data Layer: Relational database for structured data, in-memory cache for speed, vector database for intelligent search, graph database for entity relationships
- Frontend: Modern React web application with real-time updates and fluid interactions
- Deployment: Fully containerized, runs locally or in any cloud environment
- Security: Industry-standard authentication, encrypted storage, and role-based access control

1.5 Innovation Highlights

Support-First Architecture

Most AI systems route user messages to tools based on keywords. Scaffold flips this: it first assesses the user's current goal, progress stage, and friction points, then selects the right type of support -- whether that means teaching, challenging, planning, retrieving evidence, or recommending a next step. Tools are invoked only when needed, as a consequence of reasoning.

Intelligent Planning Layer

Scaffold uses a two-stage decision process: a lightweight planner analyzes what the user needs, then hands off to the main agent with only the relevant capabilities activated. This keeps responses fast, focused, and accurate.

Persistent Relationship

Scaffold maintains an evolving relationship with the user. It tracks mood, affinity, and engagement over time. As the relationship deepens, Scaffold's communication style naturally adapts -- becoming warmer, more proactive, and more personalized. This is not cosmetic; it drives retention.

Cross-Session Memory

Scaffold remembers what users are working on across sessions. When users return after a day, Scaffold picks up where they left off, surfaces unfinished work, and adjusts its greeting based on how long they have been away.

Visible AI Reasoning

Users can see what Scaffold understands, what it plans to do, and why. Every proactive action comes with approve/deny/discuss/edit controls. This transparency builds trust and gives users full control over the agent's behavior.

Safe Autonomy

Scaffold can execute tasks on the user's computer -- running commands, browsing the web, or interacting with desktop applications. Every action is auto-classified by risk level, with higher-risk operations requiring explicit user approval. Users stay in control.

1.6 Intellectual Property

Scaffold represents significant original work: over 200 source files covering the agent architecture, knowledge retrieval pipeline, emotional engine, and user interface. The support-first orchestration model

and the dual-stage planning system are novel designs with patent potential. The full codebase is proprietary.

1.7 Future Benefits

- **Productivity:** By shifting orchestration from user to agent, Scaffold reduces cognitive overhead by an estimated 40-60%.
- **Learning:** Stage-aware support creates adaptive learning pathways, potentially reducing learning time by 30-50% for complex topics.
- **Continuity:** Cross-session memory eliminates the cold-start problem, saving 10-15 minutes per session in context reconstruction.
- **Retention:** The relationship system creates emotional investment that increases with usage, driving strong long-term engagement.
- **Enterprise Readiness:** Multi-tenant architecture, role-based access, and audit logging support team and enterprise deployment.

2. Technology, Products, and Services

2.1 Product Development Strategy

Scaffold follows a strict layered development discipline. Each layer must be stable and used daily before the next one begins.

Layer 0 -- Stabilize (Current)

Ensure reliability: fix all broken flows, validate real-time streaming, write automated tests for every feature.

Layer 1 -- Daily Driver

Make Scaffold the first tool users open for work. Includes intent-aware routing (replacing simple keyword matching), auto-capture of notes from conversations, reliable task tracking, and character identity tuning.

Layer 2 -- Power Tools

Add capabilities for technical users: index and search personal codebases, stable command-line and browser automation, lightweight file editing with preview.

Layer 3 -- Memory and Continuity

Make returning to Scaffold feel like resuming, not restarting. Time-aware re-entry briefings, structured long-term memory, and graceful forgetting of outdated information.

Layer 4 -- Proactive Presence

Scaffold begins surfacing useful suggestions without being asked. Every proactive action is shown as a card with clear approve/deny/discuss controls. The user is always in control.

Layer 5 -- Situational Awareness

Scaffold adapts to the user's working style, energy level, and cognitive state. Predictive context loading, personalized behavior, and deeper collaboration calibration.

2.2 Production Strategy

Scaffold is deployed as a containerized application, enabling:

- Local-first development: the full product runs on a single machine
- Cloud-ready deployment: deployable to any major cloud provider (AWS, GCP, Azure)
- Horizontal scaling: stateless servers behind load balancers, with shared database layers
- Multi-environment support: development, staging, and production via configuration

2.3 Industry Characteristics

From Chat to Agent

The industry is shifting from passive chat interfaces to proactive agents that can plan, execute, and follow

through. Scaffold is built as an agent from the ground up.

Model Agnosticism

AI models improve every few months. Scaffold is designed to swap underlying models without code changes, keeping the product current as the field evolves.

Privacy and Control

Users demand ownership of their data. Scaffold supports local deployment, encrypted storage, and per-user data isolation by default.

2.4 Product Capabilities

Capability	What It Does	User Value
Agent Chat	Multi-mode conversation with streaming	Natural interaction, no switching
Task Tracking	Auto-extracted tasks with deadlines	Nothing falls through the cracks
Note Capture	Auto-captured insights, linked together	Knowledge compounds over time
Knowledge Base	Upload docs, search intelligently	Ask questions about your own files
Memory	Long-term recall across sessions	Scaffold remembers your context
Automation	Shell, browser, desktop control	Execute without context-switching
Relationship	Mood, affinity, engagement tracking	An AI that grows with you
Guardian	Break reminders, deadline alerts	Someone watching out for you
Dashboard	Progress, streaks, overview	Motivation and clarity at a glance

3. Market Analysis

3.1 Market Positioning

Scaffold sits at the intersection of three large and growing markets:

- AI Productivity Tools -- projected USD 47 billion by 2030
- AI Agent Platforms -- projected USD 65 billion by 2030
- AI-Powered Education -- projected USD 32 billion by 2030

Scaffold's unique position: the first AI support agent that combines the depth of a coding assistant, the structure of a project manager, and the continuity of a learning companion into a single, emotionally aware colleague.

3.2 Market Demand

Knowledge Worker Overwhelm

68% of knowledge workers struggle with the pace of work (Microsoft Work Trend Index, 2024). Managing tasks, notes, and information across disconnected tools is a major contributor. Scaffold centralizes this orchestration in the agent.

The Agent Adoption Wave

Gartner predicts that by 2028, 33% of enterprise software will include agentic AI, up from less than 1% in 2024. The market is ready for products that go beyond simple chat.

Continuous Learning Economy

In the AI era, the ability to rapidly learn and apply new skills is a career differentiator. Scaffold's study-support capabilities directly serve this growing demand.

3.3 Industry Forecast

The AI productivity tools market is expected to grow from USD 14.5 billion in 2024 to USD 47 billion by 2030 (CAGR ~21%). The agent-based segment within this market is expected to grow at 35-45% CAGR as users move from passive assistants to proactive agents.

Adoption timeline:

- 2025-2026: Early adoption by developers and AI engineers
- 2026-2027: Mainstream adoption by knowledge workers in tech, finance, research
- 2027-2028: Enterprise adoption with team features and compliance
- 2028-2030: Expansion into education, healthcare, professional services

Early movers will build user relationships and data moats (personalized memory, knowledge graphs) that late entrants cannot replicate. The window is now.

4. Competition Analysis

4.1 Competitor Landscape

Competitor	Category	Strengths	Key Gap
ChatGPT / Claude	AI Chat	Model capability, brand	No persistence or proactive support
Notion AI	Productivity	Large user base, workspace	AI is add-on, user still organizes
Cursor / Copilot	Coding AI	Deep IDE integration	Code-only, no broader support
Khanmigo	AI Tutoring	Educational content	Education-only, no work tools
AutoGPT / CrewAI	Agent Framework	Autonomous execution	Framework, not product; no UI
Mem.ai / Rewind	AI Memory	Capture and recall	Passive recall, no active guidance

4.2 Domestic vs. Overseas

International Players

OpenAI, Anthropic, and Notion lead globally, but none offers a persistent, emotionally aware agent that supports the full work-study lifecycle. Their products are either general-purpose chat or manual-first productivity tools.

Domestic Opportunity

AI-native startups in Vietnam and Southeast Asia are primarily focused on chatbot deployment and SaaS localization. This creates an open lane for a deeply engineered, agent-first product. Vietnam produces 50,000+ tech graduates annually -- a strong initial user base.

4.3 Competitive Advantages

- Depth of integration: 10 subsystems in one coherent agent, not a collection of plugins.
- Support-first design: Intervention reasoning before tool routing -- a structural advantage that is expensive to retrofit.
- Emotional moat: Relationship progression creates switching costs that grow with usage.
- Memory moat: Accumulated knowledge about user patterns, goals, and context increases value over time.
- Visible AI reasoning: Full transparency and user control over proactive actions builds trust.

5. Business Model

5.1 Marketing Strategy

Phase 1: Developer Community (Months 1-6)

- Open-source the core framework to build trust and community
- Launch on Product Hunt, Hacker News, and AI developer communities
- Publish technical content (blog, video, conference talks)
- Build a Discord community for early adopters

Phase 2: Knowledge Worker Expansion (Months 6-18)

- Launch hosted cloud version for non-technical users
- Partner with coding bootcamps, universities, and learning platforms
- Referral program with gamified incentives
- SEO and content marketing

Phase 3: Enterprise and Team (Months 18-36)

- Team features: shared knowledge bases, team tasks, admin controls
- Enterprise security: SSO, audit logging, data residency
- Sales-led growth for enterprise accounts

5.2 Profit Model

Primary: Subscription (SaaS)

Tier	Price/Month	Includes	Target
Free	USD 0	5 chats/day, basic search, 1 GB	Students, evaluators
Pro	USD 19	Unlimited, full features, 10 GB	Individual professionals
Team	USD 39/user	Shared workspace, admin panel	Small teams
Enterprise	Custom	SSO, audit, SLA, support	Organizations

Secondary Revenue

- API access for developers (usage-based pricing)
- Marketplace for community-created add-ons and templates
- AI model token passthrough at margin (users can also bring their own keys)
- Enterprise consulting for custom deployment and integration

5.3 Go-to-Market

Target user personas, in order of priority:

- Primary: AI engineers and researchers (early adopters, strong community influence)

- Secondary: Software developers (large market, tool-savvy)
- Tertiary: Graduate students and researchers (high engagement, long-term potential)
- Future: Knowledge workers in tech, finance, consulting (enterprise expansion)

The free tier is designed to demonstrate Scaffold's value within the first session. The conversion trigger is the moment users feel the relationship becoming personal -- typically after 3-5 sessions. At that point, switching costs are already real.

6. Financial Situation

6.1 Investment to Date

Category	Amount (USD)	Details
Development	~15,000	Infrastructure, API credits, tools
Annual infrastructure	~3,600	Cloud, databases, hosting
Annual AI costs	~2,400	Model inference, search, ranking
Annual tools	~1,200	Observability, design, testing
Total to date	~22,200	12 months of bootstrapped development

6.2 Revenue Projections

Year	Free Users	Paid Users	ARPU	Annual Revenue
1	2,000	100	\$19	\$22,800
2	12,000	800	\$22	\$211,200
3	50,000	5,000	\$27	\$1,620,000
4	150,000	18,000	\$32	\$6,912,000
5	400,000	50,000	\$35	\$21,000,000

Assumes 5% free-to-paid conversion (developer tools with strong community typically achieve 5-8%). Year 1 growth is organic (open-source, Product Hunt, community); paid marketing scales from Year 2 onward. ARPU increases as team and enterprise tiers gain traction.

6.3 Cost Structure

Category	Year 1	Year 2	Year 3
Team (Vietnam-based)	\$120K	\$360K	\$720K
Infrastructure	\$12K	\$48K	\$120K
AI model costs	\$8K	\$36K	\$90K
Marketing	\$12K	\$72K	\$200K
Operations	\$6K	\$24K	\$60K
Total	\$158K	\$540K	\$1,190K

Team costs reflect Vietnam market rates. Senior AI and full-stack engineers in Vietnam command USD 40,000-60,000 per year -- significantly lower than US equivalents while maintaining comparable output, supported by a strong and growing local engineering talent pool. Infrastructure and AI model costs scale with user volume.

6.4 Profit Forecast

	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	\$23K	\$211K	\$1.6M	\$6.9M	\$21M
Costs	\$158K	\$540K	\$1.2M	\$3.5M	\$8.4M
Net	-\$135K	-\$329K	+\$430K	+\$3.4M	+\$12.6M
Margin	-587%	-156%	27%	49%	60%

Path to profitability by mid Year 3. Margins are driven primarily by SaaS operating leverage -- revenue scales faster than headcount and infrastructure. These projections assume AI model costs remain flat; even under that conservative scenario, margins exceed 55% at scale due to the high gross margin of subscription software.

7. Fund Demand

7.1 Entry Amount

Total bootstrapped investment to date: approximately USD 22,200 over 12 months. The product has progressed from concept to a functional prototype with a complete agent architecture, knowledge retrieval system, emotional engine, and production frontend.

7.2 Financing Amount and Use

Seeking seed funding of USD 500,000 to USD 1,000,000 to reach market launch within 12 months.

Category	Share	Amount (USD)	Purpose
Engineering	45%	225K - 450K	Hire 2-3 senior engineers (Vietnam)
Infrastructure	20%	100K - 200K	Cloud, databases, AI costs for 18 months
Marketing	15%	75K - 150K	Community, content, launch campaigns
Operations	10%	50K - 100K	Legal, compliance, admin
Reserve	10%	50K - 100K	Contingency

Milestones

- Month 0-3: Stabilize core, launch private beta with 100 early adopters
- Month 3-6: Add power-user features, launch public beta on developer communities
- Month 6-9: Add cross-session memory, launch hosted cloud version, reach 1,000 users
- Month 9-12: Add proactive features, launch paid tiers, reach 250 paying customers
- Month 12-18: Enterprise features, Series A readiness

7.3 Return on Investment

- Break-even: Month 28-32 (mid Year 3)
- 3x return: Achievable by Year 4 through revenue growth or follow-on valuation
- 10x+ return: Achievable by Year 5 through enterprise expansion

Exit opportunities:

- Acquisition by major productivity or AI platforms (Notion, Atlassian, Microsoft, OpenAI, Anthropic)
- Independent growth to USD 100M+ ARR as a standalone SaaS company
- Technology licensing of the support-first architecture and relationship engine